

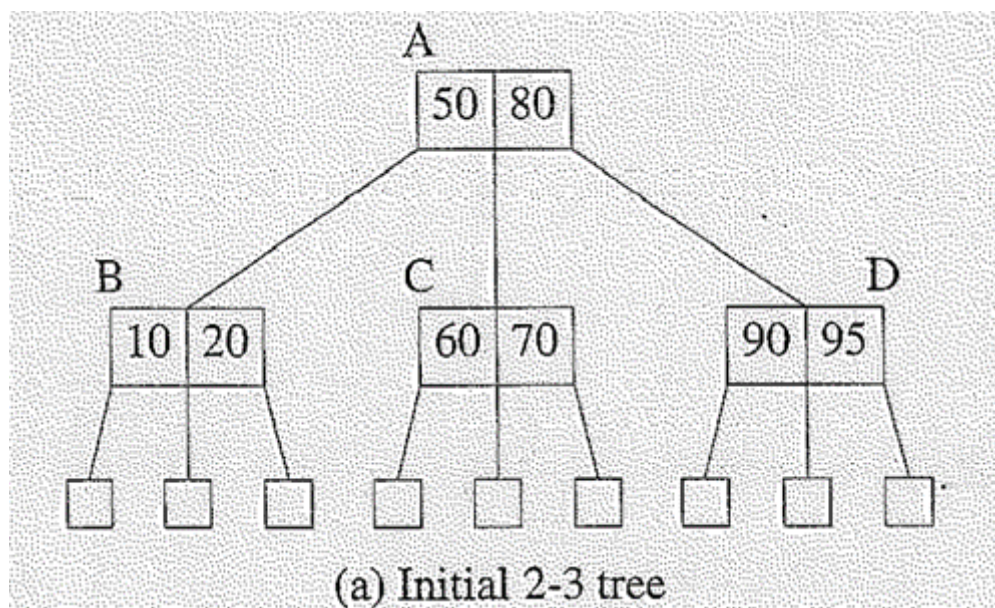
第10次作业总结

P623 题2

题目

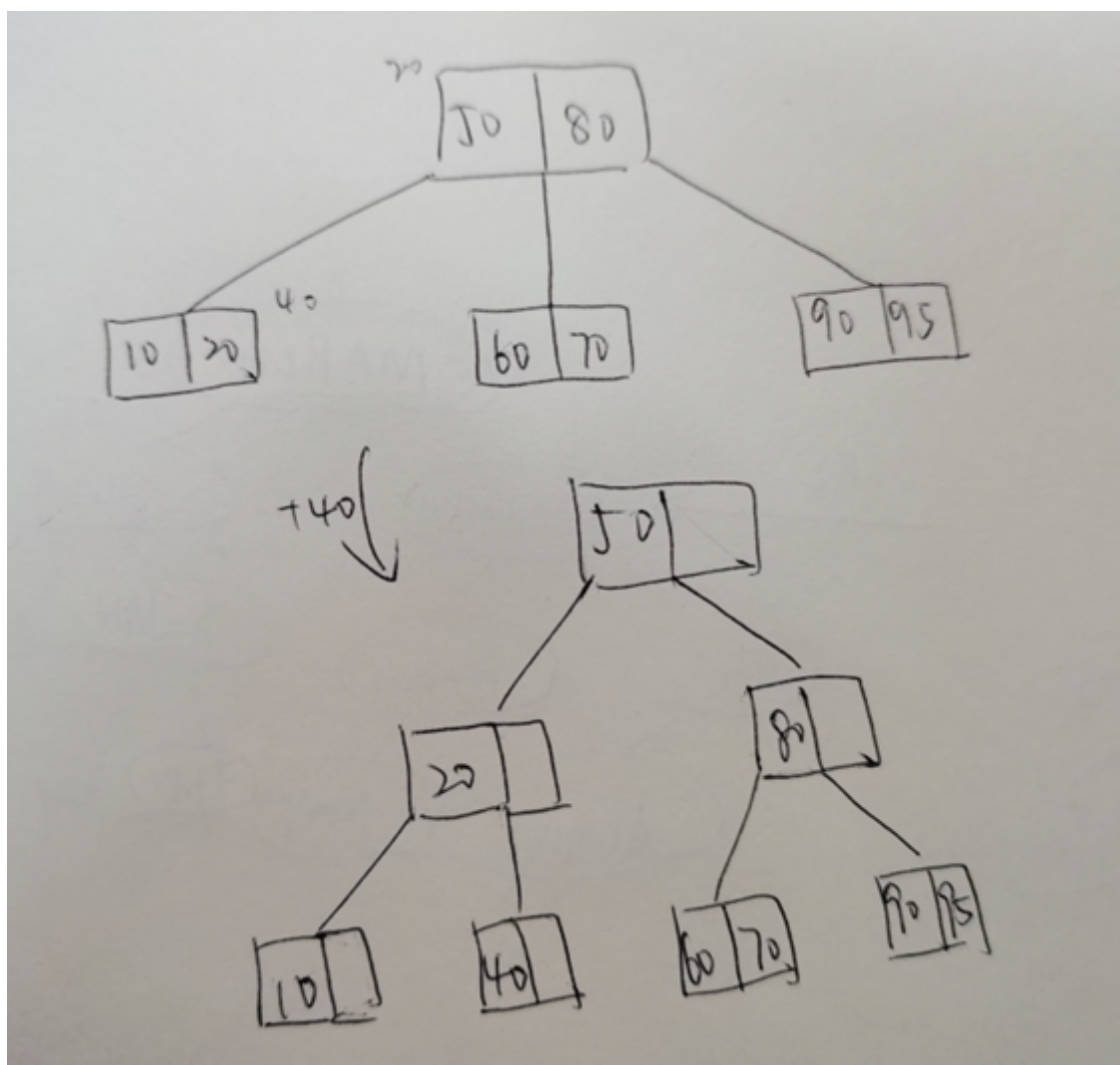
Use the insertion algorithm of Program 11.2 to insert an element with key 40 into the 2-3 tree of Figure 11. 9(a). Show the resulting 2-3 tree.

```
1  // Insert element x into a disk resident B tree
2  Search the B tree for an element E with key x.K.
3  if such an E is found, replace E with x and return;
4  Otherwise, let p be the leaf into which x is to be inserted;
5  q = NULL;
6  for(e = x; p; p = p->parent())
7  {  //(e,q) is to be inserted into p
8      Insert (e,q) into appropriate position in node p;
9      Let the resulting node have the form: n, A0, (E1, A1), ..., (En, An);
10     if(n <= m-1){ //resulting node is not too big
11         write node p to disk; return;
12     }
13     // node p has to be split
14     Let p and q be defined as in Eq.(11.5);
15     e=E_[m/2+1];
16     write nodes p and q to the disk;
17 }
18 // a new root is to be created
19 Create a new node r with format 1, root, (e,q);
20 root = r;
21 write root to disk;
22
23 // Program 11.2: Insertion into a B-tree
```



参考答案

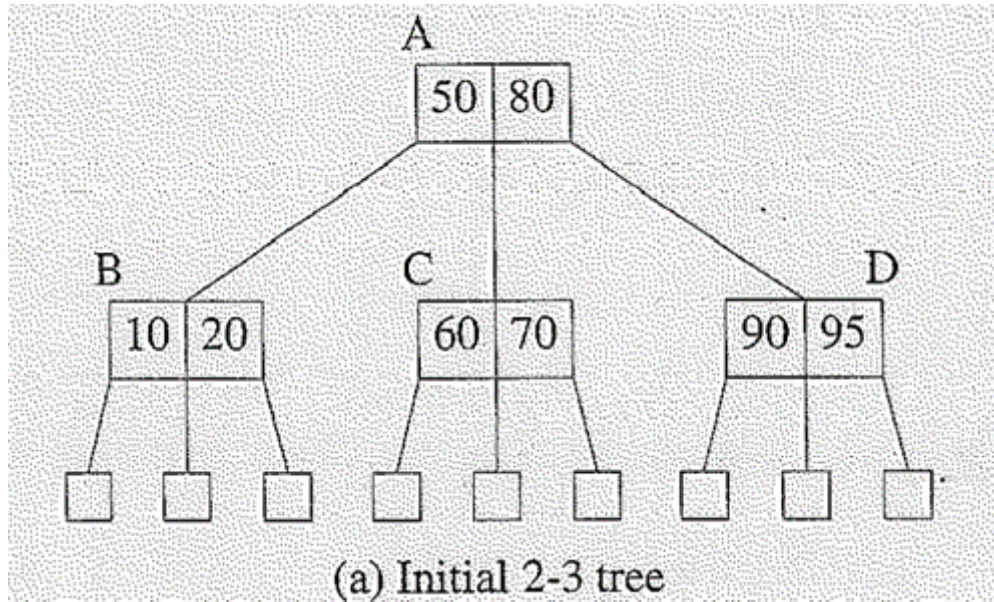
可参考如下解答：



P623 题4

题目

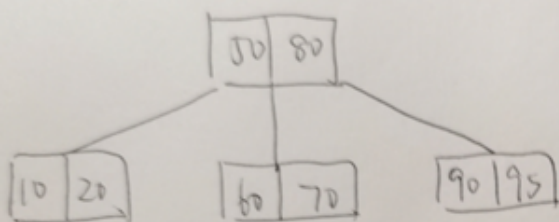
Use the deletion algorithm of Program 11.3 to delete the elements with keys 90, 95, 80, 70, 60 and 50, in this order, from the 2-3 tree of Figure 11.9(a). Show the resulting 2-3 tree following each deletion.



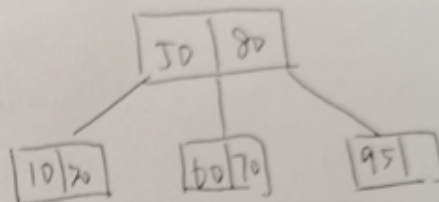
参考答案

可参考如下解答：

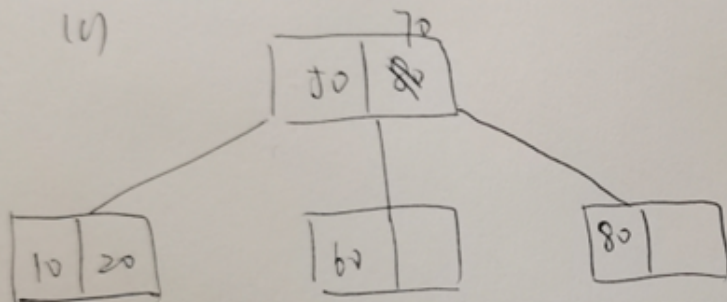
(a)



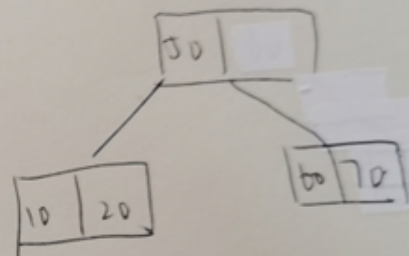
(b)



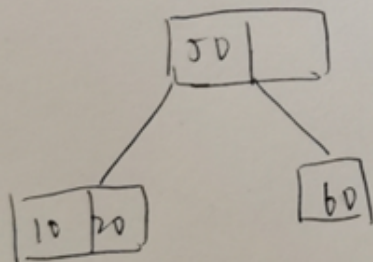
(c)



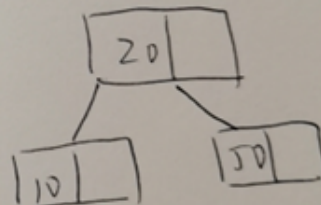
(d)



(e)



(f)



(g)

